

Algebraic Number Theory

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Disclaimer

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To separate the contents of the course to actual additions or out of context information, a black band will be added by its side like the one on this comment.

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Chapter 1

Introduction

- **Problem 1:** Find all prime numbers $p \in \mathbb{N}$, such that $\exists x, y \in \mathbb{Z}, p = x^2 + y^2$.
For small values we have

p	(x, y)
2	(1, 1)
3	✗
5	(1, 2)
7	✗
11	✗
13	(2, 3)

We have the following observation, if $p \neq 2$ then $p = x^2 + y^2$ then $p \equiv 1 \pmod{4}$, which is true since $x^2 + y^2 \equiv 0, 1, 2 \pmod{4}$ and since p is an odd prime then necessarily $p \equiv 1, 3 \pmod{4}$ hence $p \equiv 1 \pmod{4}$.

Now the question is if the other implication, if $p \equiv 1 \pmod{4}$ then is it necessary that $p = x^2 + y^2$. If we consider the equation $p = x^2 + y^2$ in the Gaussian integers $\mathbb{Z}[i]$, then we have that $p = (x + iy)(x - iy)$. $\mathbb{Z}[i]$ is a Euclidean domain (if we consider $\alpha = \beta q + r$, we can rewrite it as $|\alpha/\beta - q| = |r/\beta| < 1$, the disks with radius 1 around each points of the lattice $\mathbb{Z}[i]$ cover the whole space) and thus it is a principal ideal domain, so α factors into irreducibles. Looking at $\mathbb{F}_p^*$, we have that $-1 \in \mathbb{F}_p^*$ has order 2, if $p \equiv 1 \pmod{4}$ then there exists an element $x \in \mathbb{Z}, x^2 \equiv -1 \pmod{p}, p \mid x^2 + 1 = (x + i)(x - i)$ in $\mathbb{Z}[i]$ but $p \nmid x + i$ and $p \nmid x - i$ thus p is not irreducible in $\mathbb{Z}[i]$, thus $p = \pi \cdot \pi'$ with π irreducible, we have $p = N(p) = N(\pi \cdot \pi') = N(\pi)N(\pi') \Rightarrow N(\pi) \mid p^2$, if $N(\pi) = 1$ then $\pi \in i^{\mathbb{N}}$

which is not the case, hence $N(\pi) \neq 1$ and the same for $N(\pi')$, thus $N(\pi) = p$, since $\pi = a + bi$ then $p = N(\pi) = a^2 + b^2$.

- **Problem 2:** Find all integer solutions of the equation $y^3 = x^2 + 1$. By considering it on $\mathbb{Z}[i]$ we have the equivalent equation $y^3 = (x - i)(x + i)$, $(x - i)$ and $(x + i)$ are coprime given that if there is $\pi \in \mathbb{Z}[i]$ such that it divides them both, then it divides the difference $\pi \mid 2i \Rightarrow \pi \mid 2$, and thus from the equation $2 \mid y \Rightarrow 8 \mid y^3$ thus $x^2 \equiv -1 \pmod{8}$ which is impossible, thus $x - i$ and $x + i$ are coprime. In an UFD, if a, b coprime and $ab = \gamma^2$ then a, b are squares up to a unit, we have that $x + i = (a + bi)^3 = a^3 + 3a^2bi - 3ab^2 - b^3i = (a^3 - ab^2) + (3a^2b - b^3)i \Rightarrow x = a^3 - ab^2, 1 = 3a^2b - b^3$, from the second equation we have that necessarily $b^2 = 1$, replacing it back we have that if $b = 1$ then $3a^2 - 1 = 1 \Rightarrow a^2 = \frac{2}{3}$ which is impossible, if $b = -1$ then $a = 0$ and thus $x = 1$, therefore, the only solutions there are in this case is $(x, y) = (0, 1)$.
- **Problem 3:** Solve the Diophantine equation $y^3 = x^2 + 19$. By doing the same process as earlier, we have that $y^3 = (x + \sqrt{-19})(x - \sqrt{-19})$ in $\mathbb{Z}[\sqrt{-19}]$ which is a number ring, lets imagine that $\mathbb{Z}[\sqrt{-19}]$ has the same properties as $\mathbb{Z}[i]$, by the same way we can have that $x - \sqrt{-19}$ and $x + \sqrt{-19}$ are coprime, thus we deduce that $x + \sqrt{-19} = (a + b\sqrt{-19})^3 u$, we define the norm $N : \mathbb{Z}[\sqrt{-19}] \rightarrow \mathbb{N}, a + b\sqrt{-19} \mapsto a^2 + 19b^2$, we have that the units satisfy that $N(u) = 1$ thus we deduce $N(u) = a^2 + 19b^2 = 1$ which gives necessarily $u = \pm 1$, then we have that $x + \sqrt{-19} = (a + b\sqrt{-19})^3 = (a^3 - 3 \cdot 19ab^2) + (3a^2b - 19b^3)i$, by identification and enumerating the cases for $b = \pm 1$, we have that $3a^2 = 20$ which is impossible, and $a^2 = 6$ which is impossible too, thus there are no solutions. But notice that $7^3 = 18^2 + 19$ is a solution, which is contradictory, which comes from the fact that we assumed $\mathbb{Z}[\sqrt{-19}]$ has the properties $\mathbb{Z}[i]$, but $\mathbb{Z}[\sqrt{-19}]$ is not even a UFD since $35 = 5 \cdot 7 = (4 + \sqrt{-19})(4 - \sqrt{-19})$, we can take a better ring that contains $\mathbb{Z}[\sqrt{-19}]$, which is $\mathbb{Z}[\sqrt{-19}] \subseteq \mathbb{Z}\left[\frac{1+\sqrt{-19}}{2}\right]$ called the ring of integers of $\mathbb{Q}(\sqrt{-19})$.

Exercise 1.1: Show that this implies that the only solutions for $y^3 = x^2 + 19$ are $y = 7$ and $x = \pm 18$.

But it is not necessary in general that the ring of integers of a number field is a UFD. But, it has a unique factorization of prime ideals in rings of integers. Not all ideals are principal.

- **Problem 4:** Pell equation, let $d > 0$ consider the equation $x^2 - dy^2 = 1$, again, we take it in $\mathbb{Z}[\sqrt{d}]$, $(x + \sqrt{d}y)(x - \sqrt{d}y) = 1$, the norm $x + \sqrt{d}y \mapsto x^2 - dy^2$, thus the problem of finding solutions to the Pell equation is equivalent to finding the units of the norm, almost all, if we take for example $d = 3$, we have that $x^2 - 3y^2 = -1$ is impossible since mod 4 we obtain $x^2 + y^2 \equiv 3 \pmod{4}$, therefore in this case, we have that the set of solutions is exactly the units. Since $(2, 1)$ is a solution then $2 + \sqrt{3} \in \mathbb{Z}[\sqrt{3}]^*$ and thus it has infinitely many units so infinitely many solutions, we have that $\langle -1 \rangle \times \langle 2 + \sqrt{3} \rangle \subseteq \mathbb{Z}[\sqrt{3}]^*$, more specifically, $\langle -1 \rangle \times \langle 2 + \sqrt{3} \rangle = \mathbb{Z}[\sqrt{3}]^*$.

Theorem (Dirichlet Unit Theorem): For any quadratic ring $\mathbb{Z}[\sqrt{d}]$, with $\mathbb{Z}[\sqrt{d}]^* = \langle -1 \rangle \times \langle \varepsilon_d \rangle$ where ε_d is a fundamental unit with infinite order.

Exercise 2: Let $p \equiv 1 \pmod{4}$ prime, show that $y = \left(\frac{p-1}{2}\right)!$ satisfies $y^2 \equiv -1 \pmod{p}$.

Exercise 3:

1. Show that $\mathbb{Z}[\sqrt{-2}]$ is a Euclidean domain, $N : a + b\sqrt{-2} \mapsto a^2 + 2b^2$.
2. Find the solutions for $y^3 = x^2 + 2$.

Exercise 4: Show that for $k \in \mathbb{Z}$ the sum of two integer squares if and only if $\forall p \mid k, p \equiv 3 \pmod{4}$ then p appears with even exponent in the factorization of n .

Consider the ring $R = \mathbb{Z}\left[\frac{1+\sqrt{-19}}{2}\right] = \mathbb{Z}[\alpha]$, it is not a Euclidean domain, that is, any function $h : R \setminus \{0\} \rightarrow \mathbb{Z}_{\geq 0}$ that satisfies $\forall a, b \in R, \exists !q, r \in R, a = bq + r, r = 0 \vee h(r) < h(b)$. Suppose by contradiction that h exists, we have the minimal polynomial of α is $f(X) = X^2 - X + 5$, to find R^* we consider the norm on R as $N : R \rightarrow \mathbb{Z}, a + b\alpha \mapsto a^2 + 2ab + 5b^2, x \in R^* \Leftrightarrow N(x) = 1 \Leftrightarrow (a + b)^2 +$

$4b^2 = \pm 1 \Leftrightarrow a = \pm 1, b = 0$ thus $R^* = \{\pm 1\}$. Now take $x \in R \setminus (R^* \cup \{0\})$, with $h(x)$ minimal, look at $R/(xR)$, notice that by the Euclidean division, we get a surjection $R \twoheadrightarrow R/xR$. For any $\beta \in R, \exists q, r \in R, \beta = qx + r$, if $r = 0$ then $\bar{\beta} = 0 \in R/xR$, else if $r \neq 0$ then $h(r) < h(x)$ hence $r \in R^*, \bar{\beta} = \bar{r} \in R/xR$. Therefore, $R/xR \cong \{\bar{0}, \bar{r}\} \Rightarrow \#(R/xR) \leq 3 \Rightarrow R/xR$ is either $\mathbb{F}_2$ and $\mathbb{F}_3$, we have that $\bar{\alpha} \in R/xR, \bar{\alpha}^2 - \bar{\alpha} + 5 = 0$ but $f(X)$ is irreducible in $\mathbb{F}_3[X]$.

Chapter 2

Number Fields

Definition 1 (Number Field): A field K is said to be a number field if the extension $K/\mathbb{Q}$ is finite.

Definition 2 (Number Ring): Any subring of a number field is said to be a number ring.

Theorem 3: For any number field K , there exists a primitive element α , such that $K = \mathbb{Q}(\alpha) = \mathbb{Q}[X]/(f)$ with f the minimal polynomial of α .

Example:

1. Quadratic extensions of $\mathbb{Q}$: the primitive element is canonical, $[K : \mathbb{Q}] = 2$ then there exists a unique square free integer d such that $K = \mathbb{Q}(\sqrt{d})$.
2. Cyclotomic extensions: Let $\mathbb{Q} \subseteq \overline{\mathbb{Q}}$ an algebraic closure of $\mathbb{Q}$, fix $n \in \mathbb{N}_{>0}$, and we take the roots of unity $X^n - 1$ in $\overline{\mathbb{Q}}$, there are n of them, and μ_n is a cyclic group of order n , any generator is called a primitive n^{th} root of unity ζ_n , their minimal polynomial is called the n^{th} cyclotomic polynomial $\Phi_n \in \mathbb{Z}[X]$, that is

$$\Phi_n(X) = \prod_{\zeta_n \text{ primitive}} (X - \zeta_n) \in \overline{\mathbb{Q}}[X]$$

and we have that $\deg \Phi_n = \varphi(n) = \#(\mathbb{Z}/n\mathbb{Z})^*$.

n	$\Phi_n(X)$
1	$X - 1$

2	$X + 1$
3	$X^2 + X + 1$
4	$X^2 + 1$
5	$X^4 + X^3 + X^2 + X + 1$
6	$X^2 - X + 1$
7	$X^6 + X^5 + \dots + X + 1$
8	$X^4 + 1$
9	$X^6 + X^3 + 1$
10	$X^4 - X^3 + X^2 - X + 1$

Notice that if n is prime then $\Phi_n(X) = X^{n-1} + \dots + X + 1$, also, all the coefficients of Φ_n are ± 1 . But it is false, the first counterexample is Φ_{105} because $105 = 3 \cdot 5 \cdot 7$.

2.1. Galois Theory

Definition 4 (Galois Extension): Let K be a field extension of $\mathbb{Q}$, we say that $K/\mathbb{Q}$ is a Galois extension if the minimal polynomial of any $\alpha \in K$ splits in $K[X]$, equivalently, K is splitting field of a polynomial $P \in \mathbb{Q}[X]$.

Example:

1. Quadratic extensions are Galois.
2. Cyclotomic extensions are Galois given that it is the splitting field of $X^n - 1$.
3. $\mathbb{Q} \subseteq \mathbb{Q}(\sqrt[5]{2})$, is not Galois since $\mathbb{Q}(\sqrt[5]{2}) \subseteq \mathbb{R}$ but there are complex roots of $X^5 - 2$.

Theorem 5 (Galois Theorem): Let $K/\mathbb{Q}$ be a Galois extension, there is an inclusion reversing bijection:

$$\begin{aligned} \{\text{subfield of } K\} &\longleftrightarrow \{\text{subgroups of } \text{Gal}(K/\mathbb{Q})\} \\ L &\rightarrow \text{Gal}(K/L) \\ K^H &\leftarrow H \end{aligned}$$

where

$$\text{Gal}(K/\mathbb{Q}) = \{\sigma : K \rightarrow K \text{ automorphism} \mid \sigma|_{\mathbb{Q}} = \text{id}|_{\mathbb{Q}}\}$$

$$K^H = \{x \in K \mid \forall \sigma \in H, \sigma(x) = x\}$$

and we have that $[L : \mathbb{Q}] = (\text{Gal}(K/\mathbb{Q}) : \text{Gal}(K/L))$, $|H| = [K : K^H]$
and we have that H is normal in $\text{Gal}(K/\mathbb{Q})$ if and only if K^H is Galois.

Example:

1. Consider $K = \mathbb{Q}(\sqrt[3]{5}, \zeta_3)$, we have that $[K : \mathbb{Q}] = 6$, it is Galois since it is the splitting field of $X^3 - 5$ over $\mathbb{Q}$, we have then $\# \text{Gal}(K/\mathbb{Q}) = 6$, we compute the elements of $\text{Gal}(K/\mathbb{Q})$, we have

$$\begin{aligned} \sigma : \begin{cases} \sqrt[3]{5} \mapsto \zeta_3 \sqrt[3]{5} \\ \zeta_3 \mapsto \zeta_3 \end{cases} & \quad \tau : \begin{cases} \sqrt[3]{5} \mapsto \sqrt[3]{5} \\ \zeta_3 \mapsto \zeta_3^2 \end{cases} \\ \sigma^2 : \begin{cases} \sqrt[3]{5} \mapsto \zeta_3^2 \sqrt[3]{5} \\ \zeta_3 \mapsto \zeta_3 \end{cases} & \quad \tau^2 : \begin{cases} \sqrt[3]{5} \mapsto \zeta_3 \sqrt[3]{5} \\ \zeta_3 \mapsto \zeta_3^2 \end{cases} \\ \sigma^3 : \begin{cases} \sqrt[3]{5} \mapsto \zeta_3^3 \sqrt[3]{5} \\ \zeta_3 \mapsto \zeta_3 \end{cases} & \quad \tau^3 : \begin{cases} \sqrt[3]{5} \mapsto \zeta_3^2 \sqrt[3]{5} \\ \zeta_3 \mapsto \zeta_3^2 \end{cases} \end{aligned}$$

It is easy to calculate that $\sigma\tau \neq \tau\sigma$, thus $\text{Gal}(K/\mathbb{Q}) = S_3$.

2. **Cyclotomic Fields:** Let ζ_n be a primitive element of unity. Let $f \in \text{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q})$, then $f : \zeta_n \mapsto \zeta_n^a$ with $\gcd(a, n) = 1$, and we have that

$$\begin{aligned} \text{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) &\rightarrow (\mathbb{Z}/n\mathbb{Z})^* \\ (\zeta_n \mapsto \zeta_n^a) &\mapsto a \bmod n \end{aligned}$$

an isomorphism, for example taking $n = 7$, we have that $\text{Gal}(\mathbb{Q}(\zeta_7)/\mathbb{Q}) = (\mathbb{Z}/7\mathbb{Z})^* = C_6$, thus there are two subgroups C_2 and C_3 , which correspond to the subfields $\mathbb{Q}(\zeta_7 + \zeta_7^{-1})$ and $\mathbb{Q}(\zeta_7 + \zeta_7^2 + \zeta_7^4)$ respectively. We have that $[\mathbb{Q}(\zeta_7 + \zeta_7^2 + \zeta_7^4) : \mathbb{Q}] = 2$ thus $\mathbb{Q}(\zeta_7, \zeta_7^2 + \zeta_7^4) = \mathbb{Q}(\sqrt{d})$, to find d , we consider the action of $\zeta_n \mapsto \zeta_n^{-1}$ on $\zeta_7 + \zeta_7^2 + \zeta_7^4$ and we get $\text{Min}(\zeta_7 + \zeta_7^2 + \zeta_7^4) = (X - (\zeta_7 + \zeta_7^2 + \zeta_7^4))(X - (\zeta_7^3 + \zeta_7^5 + \zeta_7^6)) = X^2 - X + 2$ by having the heart to calculate it.

2.2. Ideals In Number Rings

Definition 6 (Operations On Ideals): Let R be a ring and I, J be ideals.

- **Addition:** $I + J = \{x + y \mid x \in I, y \in J\}$.
- **Multiplication:** $IJ = \{\sum x_i y_i \mid x_i \in I, y_i \in J\}$

Example:

- Let $R = \mathbb{Z}[\sqrt{-5}]$ and $I = (2, 1 + \sqrt{-5})$, we have

$$\begin{aligned} I^2 &= (2, 1 + \sqrt{-5})(2, 1 + \sqrt{-5}) \\ &= (4, 2(1 + \sqrt{-5}), -4 + 2\sqrt{-5}) \\ &= (4, 2 + 2\sqrt{-5}, 6) = (2) \end{aligned}$$

but I is not principal, suppose by contradiction that there exists $\alpha = a + b\sqrt{-5} \in R$ such that $I = (\alpha)$, we have that $(2) = I^2 = (\alpha)^2 = (\alpha^2)$ thus $(2) = (\alpha^2) \Rightarrow \alpha^2 = 2u$ with $u \in R^* = \{\pm 1\}$, so $\alpha^2 = \pm 2 \Rightarrow (a + b\sqrt{-5})^2 = a^2 - 5b^2 + 2ab\sqrt{-5} = \pm 2 \Rightarrow ab = 0 \Rightarrow a = 0$ or $b = 0$ which is a contradiction.

Definition 7 (Coprime Ideals): Let R be a ring and I, J be two ideals, we say that I and J are coprime if $I + J = R$.

Example:

- If $I = (x)$ and $J = (y)$ then $I + J = R \Leftrightarrow \exists a, b \in R, ax + by = 1$.
- If I, J are coprime then for any $n \geq 1$, I^n, J^n are coprime. Let I, J coprime, then $R = I + J \Rightarrow R = R^{2n} = (I + J)^{2n}$ which is equal to

$$\underbrace{I^{2n}}_{\subseteq I^n} + \dots + \underbrace{I^{n+1}J^{n-1}}_{\subseteq I^n} + \underbrace{I^nJ^n}_{\subseteq I^n + J^n} + \underbrace{I^{n-1}J^{n+1}}_{\subseteq J^n} + \dots + \underbrace{J^{2n}}_{\subseteq J^n} \subseteq I^n + J^n$$

thus $R = I^n + J^n$ so I^n, J^n are coprime.

Theorem 8: Let I, J, Z be ideals of R , I and J are coprime, suppose that $IJ = Z^n$ then we have that $I = (I + Z)^n$ and $J = (J + Z)^n$.

Proof sketch: $(I + Z)^n = I^n + I^{n-1}Z + \dots + Z^n = I(I^{n-1} + I^{n-2}Z + \dots + Z^{n-1} + J) = IR = I$.

Example: Consider $R = \mathbb{Z}[-19]$, we have $(18 + \sqrt{-19})(18 - \sqrt{-19}) = 7^3$ then $(18 + \sqrt{-19})_R (18 - \sqrt{-19})_R = (7)_R^3$, we have then

$$(18 + \sqrt{-19}) = [(18 + \sqrt{-19}) + (7)]^3 = (18 + \sqrt{-19}, 7)^3$$

which is not principal.

2.2.1. Fractional Ideals

Let R be a number ring, and $K = \text{Frac}(R)$ the field of fractions of R .

Definition 9 (Fractional Ideal): Let $I \subseteq K, I \neq \{0\}$, we say that I is fractional if it is an additive subgroup of K that satisfies:

- $\forall r \in R, rI \subseteq I$. (Ideal of R)
- $\exists x \in K^*, xI \subseteq R$.

Example:

1. Ideals of R are fractional ideals (integral ideals).
2. $\forall x \in K^*, xR = (x)_R$ is a fractional ideal (principal fractional ideals).
3. Take $R = \mathbb{Z}[\sqrt{-5}], I = R + R\frac{1+\sqrt{-5}}{2}$ is a fractional non-principal ideal.

Definition 10 (Ideal Quotient): Let I, J be two fractional ideals of R , we define the quotient of I over J as

$$I : J = \{x \in K \mid xJ \subseteq I\}$$

Proposition 11: Let I, J be two fractional ideals of R , then $I : J$ is also a fractional ideal of R .

Proof. $(I : J) \neq \{0\}$ since if we take $a \in I \setminus \{0\}$ and $b \in K^* : bJ \subseteq R$ then $abJ \subseteq aR \subseteq I \Rightarrow ab \in (I : J)$. It is clear that it is an additive subgroup of K . Take $x \in (I : J)$ we have that $xJ \subseteq I \Rightarrow rxJ \subseteq rI \subseteq I \Rightarrow r(I : J) \subseteq (I : J)$. Let $c \in K^*$ such that $cI \subseteq R$, and $d \in J \setminus \{0\}$, take $x \in (I : J), cdx \in R, dx \in I \Rightarrow cdx \in R \Rightarrow cd(I : J) \subseteq R$. Thus, $(I : J)$ is a fractional ideal. ■

Definition 12 (Inverse/Multiplier Ring): Let I be a fractional ideal, then

- Inverse ideal: $I^{-1} = R : I = \{x \in K \mid xI \subseteq R\}$
- Multiplier ring: $r(I) = I : I = \{x \in K \mid xI \subseteq I\}$

Proposition 13: Let R be a ring and I a fractional ideal of R .

- $II^{-1} \subseteq R$.
- If $I = (a)$ then $I^{-1} = (a^{-1})$ and $II^{-1} = R$.
- The multiplier ring $r(I)$ is a subring of K .

Proof.

- $\forall x \in I^{-1}, xI \subseteq R \Rightarrow I^{-1}I = II^{-1} \subseteq R$.
- Suppose $I = (a) = aR$, let $x \in I^{-1}$ then $xaR = xI \subseteq R \Leftrightarrow xaR \subseteq R \Leftrightarrow xR \subseteq a^{-1}R \Leftrightarrow x \in a^{-1}R$ and hence $I^{-1} = a^{-1}R = (a^{-1})$, also, $II^{-1} = (aR)(a^{-1}R) = aa^{-1}R = R$.
- Do it. ■

Example: Notice that in general, we do not necessarily have that $II^{-1} = R$. Consider $R = \mathbb{Z}[\sqrt{5}]$ and $I = (2, 1 + \sqrt{5})$, we claim that $II^{-1} \neq R$. Suppose that $II^{-1} = R$, we have

$$I^2 = (4, 6 + 2\sqrt{5}, 2 + 2\sqrt{5}) = (4, 2 + 2\sqrt{5}) = 2(2, 1 + \sqrt{5}) = 2I$$

A, multiplying both sides by I^{-1} we get $I = IR = (2)$ then I is principal, but we have that I is indeed not principal, contradiction, so $II^{-1} \neq R$.

Definition 14 (Invertible Ideal): A fractional ideal I is said to be invertible if $II^{-1} = R$. The set of invertible ideals is denoted $I(R)$.

Proposition 15: $I(R)$ forms a group under ideal multiplication.

2.2.2. Factoring Into Prime Ideals

Let $n \in \mathbb{Z}$, from the fundamental theorem of arithmetic, we have that n can be decomposed into $n = \prod_i p_i^{e_i}$ unique up to units and order. We can deduce that $(n) = \prod_i (p_i)^{e_i}$. The question now is by considering any ideal $I \subseteq R$, can we

decompose in a similar way the ideal I into a product $I = \prod \mathfrak{p}_i^{e_i}$ where $\mathfrak{p}_i$ are prime ideals.

Exercise 16: Let R be a number ring, and let $I \neq \{0\}$ be an ideal of R , prove that I contains a positive integer.

Let $\alpha \in I \setminus \{0\}$, given that $\alpha \in R$ then α is algebraic over $\mathbb{Q}$, then there exists $(a_i)_{i=1}^n \in \mathbb{Z}$, $a_n \alpha^n + \dots + a_1 \alpha + a_0 = 0$, moving elements around we get

$$a_n \alpha^n + \dots + a_1 \alpha = -a_0$$

we have that the LHS is in the ideal since $(\alpha) \subseteq I$, and thus $-a_0 \in I \Rightarrow |a_0| \in I$, and $|a_0|$ is a positive integer.

Let R be a number ring and $I \neq \{0\}$ an ideal of R , and take a prime ideal $\mathfrak{p}_i$ such that $I \subseteq \mathfrak{p}_i$, from the exercise we have that there exists $n \in I$ thus $n \in \mathfrak{p}_i$ but $\mathfrak{p}_i$ is prime then it contains a prime number p_i therefore $\mathfrak{p}_i \cap \mathbb{Z} = p_i \mathbb{Z}$.

Theorem 17: Every ideal I in a number ring R has finite index.

Proof. Let R be a number ring and $K = \text{Frac}(R)$ a number field, $[K : \mathbb{Q}] = t$. From the exercise, we have that there exists a positive integer $a \in I$. We have that $(a) \subseteq I$, then there exists a surjective map $R/aR \twoheadrightarrow R/I$, thus it is enough to prove that R/aR is finite.

Lemma: Let G be a group, such that any finitely generated subgroup of G is of order at most n , then G is a finite of order at most n .

Let M be a finitely generated subgroup of R as an additive group. M has no elements of finite order given that $\text{char}(R) = \text{char}(K) = 0$, thus M is a free abelian group, and since $[K : \mathbb{Q}] = t$ then $\text{rank}(M) \leq t$. Any finitely generated subgroup of R/aR is the image of a finitely generated subgroup of R by the surjection $R \twoheadrightarrow R/aR$, and thus any finitely generated subgroup of R/aR has order $a^k \leq a^t$, thus we conclude that R/aR has finite order. ■

Corollary 18: Every prime ideal of a number ring is maximal.

Proof. Let $\mathfrak{p}$ be a prime ideal of R , we have that $R/\mathfrak{p}$ is finite and an integral domain, thus $R/\mathfrak{p}$ is a field, and thus $\mathfrak{p}$ is a maximal ideal in R . ■

In general, prime ideal factorization does not hold even if I is invertible, the problem is the lack of invertibility of prime ideals, which offers two solutions:

1. Avoid non-invertible prime ideals.
2. Find rings R where every prime ideal is invertible (Dedkind rings).

Exercise 19:

1. Show that the ideals of $(18 + \sqrt{-19})$ and $(18 - \sqrt{-19})$ are coprime in $\mathbb{Z}[\sqrt{-19}]$.
2. Show that $(18 + \sqrt{-19}, 7)$ is a maximal ideal of index 7 in $\mathbb{Z}[\sqrt{-19}]$.
3. Show that if $I, J \subseteq R$ are coprime then $IJ = I \cap J$.
4. Compute $r(I)$ for $I = (2, \sqrt{5} + 1)$ in $\mathbb{Z}[\sqrt{5}]$.
5. Show that $I = (2, 1 + \sqrt{-5})$ is invertible in $\mathbb{Z}[\sqrt{-5}]$.
6. Show that any ideal $I \neq \{0\}$ of a number ring contains a positive integer.
7. Show that a number ring has only finitely many ideals I satisfying for some fixed bound B , $\#(R : I) \leq B$.

Consider now the field $K = \mathbb{Q}[X]/(f(X))$ where $f(X) = X^2 + 5$, and the ring $R = \mathbb{Z}[\sqrt{-5}]$, take the ideal $I = (6) = 6\mathbb{Z}[\sqrt{-5}]$ which we want to factorize, we start by finding all prime ideals $\mathfrak{p}$ such that $(6) \subseteq \mathfrak{p} \subseteq \mathbb{Z}[\sqrt{-5}]$. We have that $\mathbb{Z}[\sqrt{-5}] = \mathbb{Z} + \mathbb{Z}\sqrt{-5}$,

$$R/I = \frac{\mathbb{Z} + \mathbb{Z}\sqrt{-5}}{6\mathbb{Z} + 6\mathbb{Z}\sqrt{-5}} \cong (\mathbb{Z}/6\mathbb{Z}) \times (\mathbb{Z}/6\mathbb{Z})$$

and therefore $\# R/I = 36 = 2^2 \cdot 3^2$, since any ideal of a number field is maximal, then R/I is a field, we have 4 possible fields it can be, either a field of order 2, 3, 2^2 , 3^2 so R/I can have either characteristic 2, 3. Suppose that the characteristic is 3, and consider the map from $\mathbb{Z}[\sqrt{-5}]$ to $\mathbb{F}_3$ defined as $\varphi_3 : a + b\sqrt{-5} \mapsto a + b\alpha \pmod{3}$